

Executive Summary

We found **limited direct analogues** to an HF-alkylation plant for isobutyl acetate on Jurong Island; accordingly, we have compiled a list of **closest-match projects** (small acid/ester units and larger alkylation units) in Singapore and the region for benchmarking. Key examples include a new alkoxide plant and specialty rubber/phenol plants in Singapore, and several Chinese alkylation units (using ionic-liquid catalysts) at refineries. For each we list location, capacity, year, CAPEX (if known), and **similarity to iBAP** (0–5). These serve to **contextualize site/aux/off-site costs** and to develop factors. A summary table compares all examples. We also note data gaps (e.g. proprietary CAPEX) and suggest next steps (e.g. EPC contacts, tender sources). A mermaid flowchart illustrates using these examples to calibrate cost factors.

Historical Project Examples

We found the following relevant projects (ranked by relevance):

- **Evonik Alkoxides Plant – Singapore (Jurong Island, 2025)**

Process: Production of sodium methylate (“alkoxides”) via methanolysis of FAME (no HF). *Capacity:* 100,000 tpa ¹ ². *CAPEX:* US\$60 million ². *Site:* ~600 m² area ³ under JTC lease. *Similarity (Score 1/5):* No HF, but specialty chemical plant with corrosives and strong caustics. *Sources:* Evonik/Straits Times ² ¹.

- **Lanxess Butyl Rubber Plant – Singapore (Jurong Island, 2013)**

Process: Halobutyl rubber (isobutene polymerization with bromination). *Capacity:* 100,000 tpa ⁴. *CAPEX:* ~€400 million ⁴. *Site:* Jurong Island, likely JTC lease; employs pressurized halogenation. *Similarity (Score 0/5):* No acid catalyst (uses bromine) and high-end polymer process. *Source:* Process Worldwide ⁴.

- **INEOS (ex-Mitsui) Phenol/Cumene Complex – Singapore (Jurong Island, 2023)**

Process: Cumene oxidation to phenol (sulfuric acid catalyst). *Capacity:* 410 ktpa cumene, 310 ktpa phenol, 185 ktpa acetone ⁵. *CAPEX:* US\$330 million (acquisition price) ⁵. *Similarity (Score 0/5):* Different chemistry (H₂SO₄-catalysis), but large Jurong Island plant. *Source:* IChemE News ⁵.

- **ExxonMobil Butyl/Resins Expansion – Singapore (Jurong Island, 2018)**

Process: Halobutyl rubber and hydrogenated resins (halogenation of isobutene, hydrogenation of polymers). *Capacity:* 140,000 tpa butyl rubber (premium Halobutyl) ⁶. *CAPEX:* multi-billion USD (project described as “multi-billion dollar expansion” ⁷). *Similarity (Score 0/5):* No acid catalyst; major refinery-oriented chemical plant. *Sources:* ExxonMobil press release ⁶ ⁷.

- **Sinopec Wuhan Refinery Ionikylation – China (Wuhan, 2020)**

Process: HF-alkylation unit retrofit to composite **ionic-liquid (IL) alkylation** at 161 kbpd refinery. *Capacity:* 7,500 bbl/d (~119 ktpa) ⁸. *CAPEX:* unspecified (brownfield conversion). *Similarity (Score 2/5):* Alkylation chemistry (but IL vs HF) at refinery scale. *Sources:* Focus on Catalysts journal ⁸.

• **Sinopec Jiujiang Refinery Ionikylation – China (Jiujiang, 2019)**

Process: Similar IL-alkylation retrofit at Jiujiang refinery. *Capacity:* ~7,500 bbl/d (~119 ktpa) ⁹ .

Similarity (Score 2/5): Same technology as Wuhan. *Source:* Focus on Catalysts ⁹ .

• **PetroChina Harbin Refinery Ionikylation – China (Harbin, 2018)**

Process: IL-alkylation unit added to Harbin refinery. *Capacity:* 3,000 bbl/d (~48 ktpa) ¹⁰ . *Similarity*

(Score 2/5): Another IL-alkylation example. *Source:* Focus on Catalysts ¹⁰ .

• **PetroChina Golmud Refinery Ionikylation – China (Golmud, 2019)**

Process: IL-alkylation at Golmud city refinery. *Capacity:* 1,000 bbl/d (~16 ktpa) ¹⁰ . *Similarity (Score 2/5):*

Same IL tech at small scale. *Source:* Focus on Catalysts ¹⁰ .

We found **eight** closely relevant projects. (Few Singapore-specific HF-alkylation units exist, so we included refinery alkylations and related processes as analogues.) Other examples like oleochemical or polymer plants (e.g. acrylics, urethanes) in the region exist but are less similar in chemistry.

Similarity Assessment

Each example is scored 0–5 on similarity to an iBAP (HF-catalyzed alkylation) plant:

- *Score 5:* Direct HF alkylation with similar feedstocks (no such Asia project found).
- *Score 3–4:* Acid-catalyzed alkylations (SAC or HF) of olefins. None found in Singapore.
- *Score 2:* Alkylations using ionic liquids (Sinopec/PetroChina examples) – similar end-goal (high-octane alkylate) but different catalysts ⁸ .
- *Score 1:* Acidic specialty plants (Evonik alkoxide, butyl rubber) – involve corrosives/strong catalysts but not same reaction.
- *Score 0:* Largely dissimilar chemical processes (phenol via cumene, halogenations).

These scores guide how we **scale cost insights**: higher scores imply more relevant cost structure (e.g. materials, safety).

Example Data Table

Project (Owner)	Location (Country)	Year	Capacity	CAPEX	Sim.	Source(s)
Evonik Alkoxides Plant	Jurong Island, SG	2025	100,000 t/y (NaOCH ₃)	~US\$60M ²	1	² ¹
Lanxess Butyl Rubber (Halobutyl)	Jurong Island, SG	2013	100,000 t/y (rubber)	~€400M ⁴	0	⁴
INEOS (Mitsui) Phenol/Cumene Complex	Jurong Island, SG	2023	410k t/y cumene; 310k t/y phenol	US\$330M (acquisition) ⁵	0	⁵

Project (Owner)	Location (Country)	Year	Capacity	CAPEX	Sim.	Source(s)
ExxonMobil Butyl/Resin Plants	Jurong Island, SG	2018	140,000 t/y butyl 90,000 t/y resins	<i>Multi-billion</i> (unspecified) 6 7	0	6 7
Sinopec Wuhan Refinery (Ionikylation)	Wuhan, China	2020	7,500 bbl/d (~119 kt/y)	–	2	8
Sinopec Jiujiang Refinery (Ionikylation)	Jiujiang, China	2019	7,500 bbl/d (~119 kt/y)	–	2	9
PetroChina Harbin Refinery (Ionikylation)	Harbin, China	2018	3,000 bbl/d (~48 kt/y)	–	2	10
PetroChina Golmud Refinery (Ionikylation)	Golmud, China	2019	1,000 bbl/d (~16 kt/y)	–	2	10

Notes: Capacity in tonnes per annum (t/y) or barrels per day (bbl/d) as noted. CAPEX is listed if reported; “–” indicates not public. Sim. = similarity score.

Using Historical Data for Cost Adjustment

The flowchart below illustrates how to use these historical case data to refine your cost estimates:

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flowchart LR
    Hist["Historical Plant Data (site/aux/off % of CAPEX)"] --> Extract["Extract site/aux/off cost ratios"]
    Extract --> Adjust["Adjust estimation factors (p_S, p_A, p_0)"]
    Adjust --> Estimate["Apply to new iBAP CAPEX for site/aux/off estimates"]

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For example, if Evonik’s small alkoxide plant (low sim.) had a site development cost of X% of its (US\$60M) CAPEX, that suggests a baseline for a high-tech specialty plant. The refinery alkylation units give guidance on auxiliary equipment scaling. Use these ratios (weighted by similarity) to set default percentages, then apply to your project CAPEX.

Data Gaps & Next Steps

Gaps: We lacked public data on true HF-alkylation projects (none found in SG) and on equipment lists. CAPEX often undisclosed. Key missing data: small esterification/HF-alkylation plant CAPEX, site areas.

Next Steps:

- **Contact EPC firms:** Technip Energies, Fluor, Samsung Eng, JGC, Sembcorp, Keppel, Chiyoda have done Asian chemical plants. Inquire about similar project metrics.
- **Public Tenders/Records:** Check Government portals (e.g. geBIZ for awards, NEA/URA planning approvals for major plant expansions).
- **Industry Databases:** Subscription databases (ICIS, Wood Mackenzie) may list anonymous project figures.
- **Academic/Trade Journals:** Continue monitoring IChemE, Hydrocarbon Processing for case studies.
- **EPC Project Lists:** Some EPC contractors (e.g. Chiyoda's PDF) list capacities (see above).

These steps may uncover proprietary numbers or analogues. Given Singapore's focus on sustainable/bio-based chemistry, also look for new bio-ester or IL-based refinery projects (as proxies if HF is phased out).

Sources: We prioritized Singapore government and industry press releases, technical journals, and credible news (see citations). Many specific CAPEX figures are proprietary; we note them as "unspecified" where not found.

1 Evonik Celebrates Opening of World-Scale Alkoxides Plant in Singapore

<https://www.evonik.com/en/news/press-releases/2025/08/Evonik-Celebrates-Opening-of-World-Scale-Alkoxides-plant-in-Singapore.html>

2 3 Evonik opens its first Asia speciality chemical alkoxide plant in Singapore, eyes regional expansion | JTC | Jurong Island

<https://www.jtc.gov.sg/jurongisland/story-articles/evonik-opens-on-jurong-island---new>

4 Lanxess opens Asia's most advanced butyl rubber plant

<https://www.process-worldwide.com/lanxess-opens-asias-most-advanced-butyl-rubber-plant-a-408066/>

5 Ineos steps into Asia phenol with purchase of Mitsui's Singapore plants - News - The Chemical Engineer

<https://www.thechemicalengineer.com/news/ineos-steps-into-asia-phenol-with-purchase-of-mitsui-s-singapore-plants/>

6 7 ExxonMobil Singapore Butyl and Resins Plants Begin Production :: Exxon Mobil Corporation (XOM)

<https://investor.exxonmobil.com/company-information/press-releases/detail/252/exxonmobil-singapore-butyl-and-resins-plants-begin>

8 9 10 Sinopec starts up new IL alkylation unit at Wuhan refinery - PMC

<https://pmc.ncbi.nlm.nih.gov/articles/PMC7243807/>